

Thien-An Tran

(206) 883-0263 | thienanthedrmer@gmail.com | [GitHub](#)

EDUCATION

University of Washington | BSc. Computer Science

2022 – 2026

Relevant Coursework: Cryptography, Secure Coding Principles, Data Structures & Algorithms, Machine Learning, AI & Knowledge, Machine Organization, Database Systems Design, Game Simulation and Design

TECHNICAL SKILLS

Languages | Java, Python, JavaScript, C#, C++, SQL, C, Erlang, Assembly

Technologies | Cryptography (SHA-3, SHAKE, NIST FIPS 202, Argon2, ECIES, Schnorr), Microsoft Azure (Blob Storage), PyTorch, OpenCV, NumPy, React, Git, Node.js, Firebase, Heroku, Bootstrap, Phaser, Unity Game Engine

PROFESSIONAL EXPERIENCE

Full Stack Developer | Vietnamese Martyr's Parish

2024 – 2026

- Developed a secure document management system using React and Microsoft Azure Blob Storage
- Implemented access controls and data integrity measures for 500+ sensitive user records
- Designed and maintain a SQL database of 500+ user records with optimized queries and strict data integrity enforcement for sensitive personal documentation
- Implemented security best practices including parameterized queries, access controls, and privacy-compliant data handling for records management

Web Developer | PROstrada

2024 – 2025

- Designed and developed interactive web applications in React by translating client requirements into functional software, ensuring performance and responsiveness standards were met
- Partnered with non-technical stakeholders to translate operational requirements into software designs, aligning technical solutions with business and user needs
- Maintained and enhanced software products iteratively based on stakeholder feedback
- Managed requirements changes throughout the development lifecycle

PROJECTS

Cryptography Library | *Java*

2026

- Implemented SHA-3-256, SHA-3-512, SHAKE-128, and SHAKE-256 from scratch in Java conforming to NIST FIPS 202, verified against official test vectors
- Designed symmetric encryption and decryption system using SHAKE-128 as a stream cipher with a 128-bit random nonce and passphrase-derived key, and a SHAKE-based message authentication code for tamper detection
- Implemented NUMS ed-256-mers* Edwards elliptic curve arithmetic in Java including point addition, scalar multiplication, and subgroup validation
- Built ECIES public-key encryption scheme using SHAKE-based KDF, stream cipher, and MAC tag verification for authenticated encryption
- Implemented Schnorr digital signatures with deterministic nonce generation for message signing and verification

Secure Data Entry System | *Java, Python*

2026

- Co-developed hardened data-entry applications in both Java and Python implementing input validation, integer overflow detection, salted password hashing, and structured audit logging
- Designed Java security architecture using SHA-256 with SecureRandom salt and overflow-safe arithmetic; Python version uses Argon2 memory-hard hashing
- Implemented file path sanitization using regex whitelisting to prevent directory traversal attacks, and parameterized input handling to prevent injection vulnerabilities
- Built 20+ unit tests in Python unit test framework covering boundary conditions, overflow cases, error paths, and password verification flows
- Participated in adversarial red-team competition identifying and exploiting vulnerabilities in peer implementations while defending own codebase against attack

AI Autonomous Agent | *Python*

2025

- Designed and implemented a multi-agent autonomous navigation system in Python integrating adversarial search with probabilistic Bayesian reasoning under sensor uncertainty
- Implemented and documented five pathfinding algorithms (BFS, DFS, UCS, Greedy, A*) analyzing optimality and efficiency tradeoffs across constrained grid environments
- Built a Bayesian probabilistic reasoning engine handling false-positive and false-negative sensor readings, maintaining belief distributions updated via posterior inference

Calculrush | *Node.js, Phaser, Socket.io*

2024

- Built real-time multiplayer educational game with server-authoritative Node.js and Socket.IO, deployed live with 25+ concurrent users
- Designed client-server event protocol enabling parallel frontend and backend development across a three-person team

Vicinity | *JavaScript, React, Firebase, Google Maps API*

2023

- Collaborated with a team to build a social event-discovery app with real-time Firebase data retrieval and Google Maps API radius filtering
- Implemented Firestore queries for category, date, and location filtering enabling users to find local events within a custom radius